

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. (CE/IT/EC/CSE) Examination

May 2018

CE142 / CE104 Object Oriented Programming with C++**Date: 05.05.2018, Saturday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION 1**Q-1 Do as Directed.****[07]**

- a. Suppose 'ob1' and 'ob2' are the objects of class 'Copy', *printX()* (prints value of X) and *setX()* (sets value of X) are public member functions of same class. 'int *X' is private data member of class Copy. All the required constructors have been provided accordingly. Consider following statements.
Copy ob1(10);
Copy ob2=ob1;
ob1.setX(12);
ob2.printX();
The output of last statement is _____.
- b. The _____ operator is used to identify the class to which a member function belongs.
- c. **Choose the appropriate option:** The 'new' operator without 'delete' causes the memory loss which is known as _____.
(Memory Leak / Dangling pointer / Heap memory loss)
- d. **State True or False:** Consider the statement 'float a=b=12.34;' Here, value of 'a' is 12.34.
- e. **State True or False:** 'const' is keyword but it is not 'qualifier'.
- f. **State True or False :** Polymorphism can be achieved by C++ structure.
- g. **State True or False:** C++ does not provide garbage collection facility.

Q-2 Answer the following questions. [Any Two]**[14]**

1. List any three differences between procedural programming paradigm and object oriented programming paradigm. Explain the terms encapsulation, data abstraction, dynamic binding and polymorphism as the pillars of object oriented concepts along with its programming syntax.
2. Explain 'friend function' along with its characteristics and example.
Draw the memory diagram for *three* objects for following private data members and public member function of class 'number'. *Note that 'friend void display (number) is not part of class 'number'.*

```
int a ; static float b; const char c;
void getdata_a();
static void getdata_b();
friend void display(number);
```

3. Define constructor and destructor. List any three characteristics of constructor. Justify following sentences along with the arguments.
 - a. Constructor can be overloaded but destructor cannot be overloaded.
 - b. A reference variable is used as argument to the copy constructor. We cannot pass the argument by value to a copy constructor.

Q-3 Attempt following programs .[Any two]

[14]

1. Create a class 'mobile' having data members *year_of_manufacture* , *price*. The class has member function *get()* and *print()*. Define a member function *bool validity()* to check the validity.(A mobile is invalid if it is older than 2005).Input data for four mobiles in *main()* and display information about the phones which are invalid. (*Use array of objects*)
2. Create a function *void calculator(char a, int b, int c)* where 'int b' and 'int c' have default arguments 50 and 100 respectively. 'char a' could be '+', '-' or '='. Mention switch statement inside function *calculator()* to perform '+', '-' and '=' between 'b' and 'c' variables. Create another function *float calculator(float pi, float r)* which will print area of circle for the value of 'r'. Call both the function from *main()* and print necessary values.
3. Create a class 'reverse' which will have *int num* as private data member and *bool check_palindrom()* as public member function. Create default constructor, parameterized constructor and copy constructor. Find whether the 'num' is palindrome or not and print in *main()*. The number is palindrome when number and its reverse number are same. For ex, 121 is palindrome number.

SECTION – II

Q-4 Do as Directed.

[07]

- a. '.' is also known as _____ operator which cannot be overloaded.
- b.


```
class A{
    public : class B {
        public : B(int i): data(i){
        }
        int data;
    };
    class C: public A{
        class D:public A::B{ };
    };
    This is _____ type of inheritance.
```
- c. The _____ pointer is used in C++ to implicitly bind a member function call with the calling object.
- d. The *seekg()* function moves the associated file's _____ pointer while

- the seekp() function moves the associated file's _____ pointer.
- e. **State True or False:** Every class with virtual function points to VTABLE using its VPTR, where VTABLE contains execution sequence of virtual functions.
 - f. **State True or False:** In multipath inheritance, in order to remove duplicate set of records in child class, we write virtual function in base class.
 - g. **State True or False:** setfill() is an ios format function to specify a character that is used to fill the unused portion of field.

Q-5 Attempt following questions. [Any Two] [14]

1. Define inheritance. Explain any three types of inheritance with its diagram and programming example.
2. List any three differences between static binding and dynamic binding. Explain how virtual functions are useful to achieve run-time polymorphism with example.
3. Explain hierarchy of C++ stream classes. Narrate and explain any four ios format function in detail with example.

Q-6 Attempt following programs using C++ language concepts. [Any two] [14]

1. Create a class 'student' which has private *int roll_num* , public void *getnum()* void *putnum()*. Create another class 'test' which inherits class 'student' and have private data members *int marks1* , *int marks2*, public member functions void *getmarks()* and void *putmarks()*. The class 'sports' inherits class 'student' and have private data member *int score* , public member function void *getscore()* and void *putscore()*. The class 'results' publically inherits class 'test' and 'sports' and prints *roll_num* and *average* of *marks1,marks2* and *score*. Create object of class 'result' and call appropriate function from main() to print necessary details.
2. Create a class 'Measure' having members: *int meter* and *int cm*. The class has void *get()* and void *put()* functions. Overload operator '+' and '-' such that they support $M1=M2+15$ and $M3=M1 - 4.5$. Also overload '+' and '-' such that they support $M1=5.0+M2$ and $M3=2.0 - M4$. Write a main() to test the class.
3. Create a class 'inventory' which has private data members *char name[10]*, *int code*, *float cost* , public member function void *readdata()* and void *writedata()*. Declare three objects for 'inventory'. Open the file 'stock.dat' and write the details about three objects in file and read the same details from file using I/O binary files operation.
